

AHZOTO

instrumentation & meters

OPERATING MANUAL

EN



VARIABLE FREQUENCY DRIVE ZT-D12 SERIES

CONTENTS

Suitable for driving single-phase and three-phase motors

- Maximum rated current of single-phase motor 10A
- Single phase submersible pump/single-phase deepwater pump
- Three phase induction motor/three-phase asynchronous motor
- Start/Stop, Forward/Reverse, Jog, Multi-stage speed, UP/DOWN frequency give

INTRODUCTION

- Thank you for using our D12 series inverter, which is a high-quality, multifunctional, and low-noise universal vector control product.
- This user manual introduces the size and functional parameter settings of the D12 series frequency converter. Please read this manual carefully before use.
- This manual is an attachment that comes with the machine. Please keep it in a safe place for future use during repairs and maintenance.
- The product can only be used for motors, not for driving household appliances.

PRODUCT SELECTION

Suitable for single-phase motors: The maximum current cannot exceed 10A. Single phase single capacitor, single-phase submersible pump, single-phase deepwater pump, etc.

Single-phase input AC 220V(200V-240V) Single-phase output AC 220V(0-240V)

Model	Power	Output current
D12-0.4	0.4KW	2.5A
D12-0.75	0.75KW	5A
D12-1.5	1.5KW	7A
D12-2.2	2.2KW	10A

This type is suitable for single-phase motors, single-phase single capacitor motors, single-phase submersible pumps, and single-phase deepwater pumps. Please choose the appropriate power based on the rated current on your motor nameplate. For example, if the rated current on the motor nameplate is 9A, 2.2KW needs to be selected. Note that if it exceeds 10A, it cannot be used. It is not recommended to use dual capacitor motors because the failure rate is particularly high.

Suitable for three-phase motors:

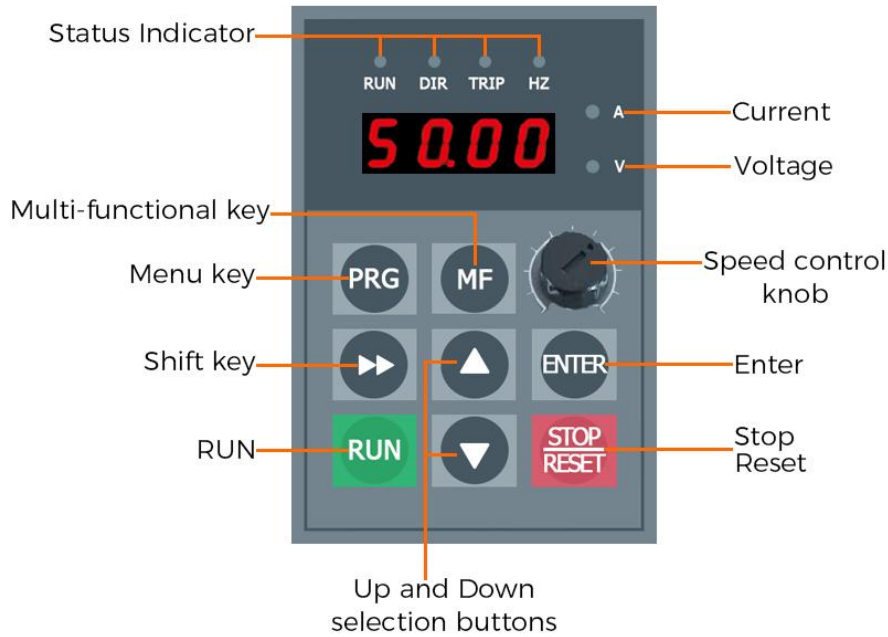
Three-phase induction motor, three-phase asynchronous motor.

Single-phase input AC 220V(200V-240V) Single-phase output AC 220V(0-240V)

Model	Power	Output current
D12-0.4G1	0.4KW	2.5A
D12-0.75G1	0.75KW	5A
D12-1.5G1	1.5KW	7A
D12-2.2G1	2.2KW	10A

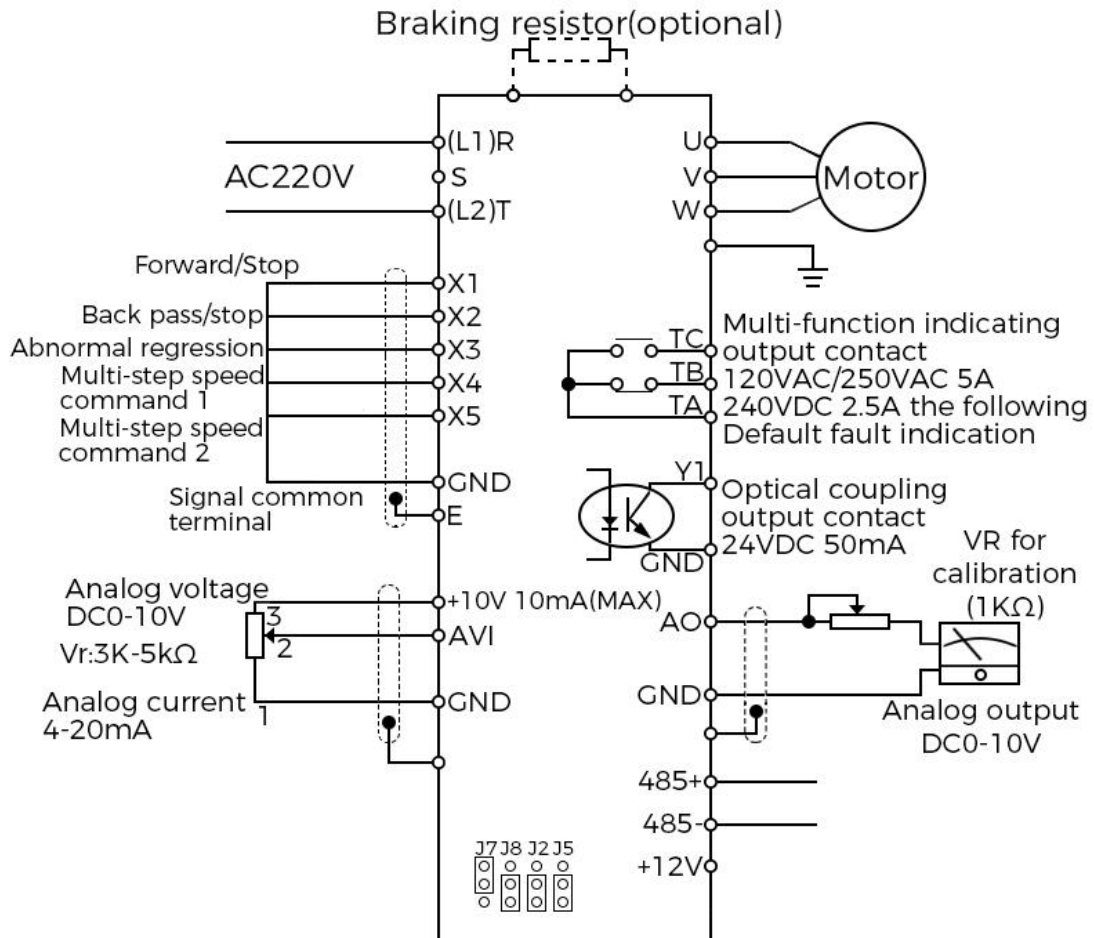
Please select the appropriate model and power based on the voltage, power (hp), and rated current on the motor nameplate. To better match your motor, please chat with me and I will help you choose the appropriate type

CONTROL PANEL



The operation panel is detachable and can be remotely operated using an extension cable.
(Extension cable needs to be purchased separately)

MAIN CIRCUIT WIRING DIAGRAM

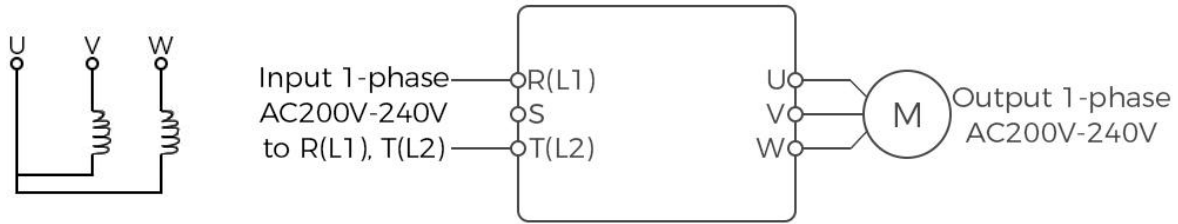


J7 is selected as the ground terminal and is disconnected by default
J8 is selected as the 485 terminal resistor and is disconnected by default
J2 is the AO output selection, default voltage output
J5 is the AI input selection, default voltage input

WIRING DIAGRAM

For the selection of single-phase motors: It is not recommended to use single-phase dual capacitor motors as they have a high failure rate. Single phase single capacitor motor, single-phase submersible pump, single-phase deepwater pump, please select the appropriate power according to the rated current on the motor nameplate. For example, if the rated current on the motor nameplate is 7.5A, 2.2KW needs to be selected. If the rated current exceeds 10A, it cannot be used. Because our maximum power is 2.2KW, with an output current of 10A

Input 1ph AC220V(200V-240V) Output 1ph AC220V(AC0-240V)

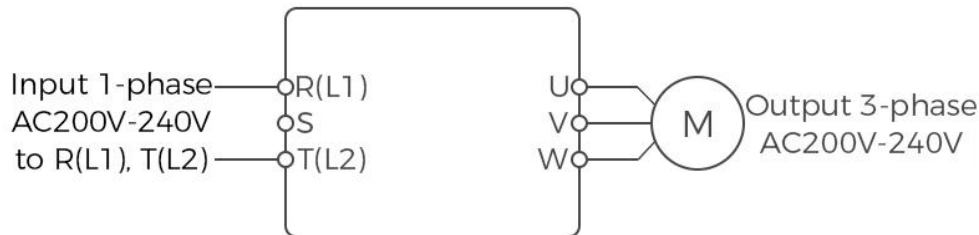


Before wiring a single-phase motor, please remove the capacitor (dual capacitors are not recommended as they are prone to malfunctions)

1. The input is connected to L1/R and L2/T, and the output UVW is connected to your motor. The UVW of VFD is directly connected to your motor, and it is prohibited to add contactors or other switching devices in the middle,

2. For the wiring of single-phase single capacitor motors, please remove the capacitor before use and then connect it to UVW. Of course, you can take pictures of your motor junction box and send them to us, and we will teach you how to wire based on your pictures.

Input 1ph AC220V(200V-240V) Output 3ph AC220V(AC0-240V)



1. The input is connected to L1/R and L2/T, and the output UVW is connected to your motor. The UVW of VFD is directly connected to your motor, and it is prohibited to add contactors or other switching devices in the middle.

Parameter Description

○— Parameters that can be modified in any state.

×— Parameters that cannot be modified in the running state.

◆— The actual detection parameters cannot be modified.

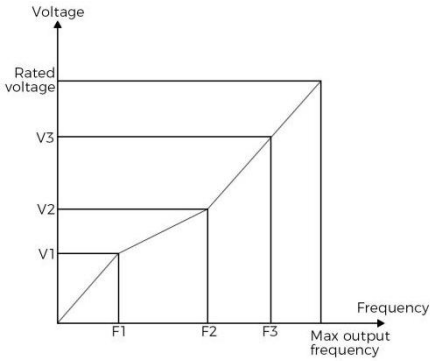
◇— The manufacturer parameters are limited to the manufacturer and are prohibited from being modified by the user.

P0 group - Basic operating parameters					
Function code	Name	Content	Range	Factory setting	Change
P0.00	VFD power specifications	Display current power	0.10~99.99KW	Model setting	◆
P0.01	Main controller software version	Display the current software version number	1.00~99.99	1.00~99.99	◆
P0.02	Run command channel selection	0: panel run command channel	0~2	0	○
		1: terminal running command channel			
		2: Communication running command channel			
P0.03	Frequency given selection	0: panel potentiometer	0~8	0	○
		1: Number given 1, adjust the ▲ and ▼ keys on the control panel			
		2: Number given 2, terminal UP/DOWN adjustment			
		3: AVI simulation given (0-10V/0-20mA)			
		4: Combination given			
		5: Retain			
		6: Communication given			
		7: Retain			
		8: MPPT given, note: When selecting a combination given, the combination given method is selected in P1.15.			
P0.04	Maximum output frequency	The maximum output frequency is the highest frequency allowed by the frequency inverter and is the reference for the acceleration/deceleration setting.	MAX{50.0, [P0.05]}-999.9Hz	50.0Hz	X
P0.05	Upper limit frequency	The operating frequency cannot exceed this frequency	MAX{0.01.0, [P0.06]}-[P0.04]	50.0Hz	X
P0.06	Lower frequency limit	The operating frequency cannot be lower than this frequency	0~Upper limit frequency	0.0Hz	X
P0.07	Processing when the lower limit frequency is reached	0: Zero speed operation	0~2	0	X
		1: Operate at the lower limit frequency			
		2: Shutdown			

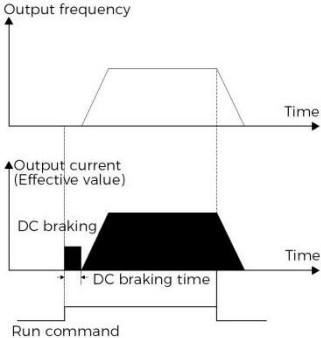
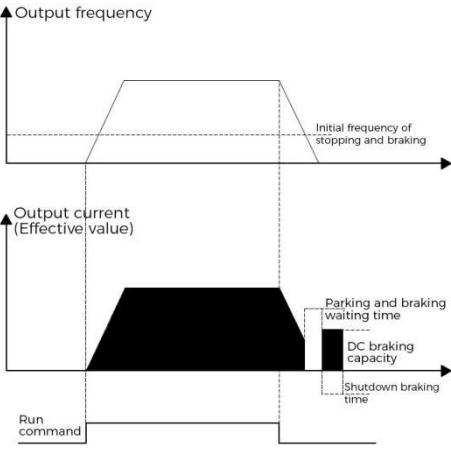
P0 group - Basic operating parameters

Function code	Name	Content	Range	Factory setting	Change
P0.08	Operating frequency digital setting	The set value is the initial value given by the frequency number	0.0~Upper limit frequency	10.0Hz	○
P0.09	Digital frequency control	LED ones place: power down storage 0: Storage, 1: No storage	0000~2111	0000	○
		LED tens: stoptokeep 0: Keep, 1: don't keep			
		LED hundreds place: UP/DOWN Negative Frequency Adjustment 0:invalid, 1:P0.03+PID			
P0.10	Accelerate time	Time required for VFD to accelerate from 0 to maximum output frequency	0.1~999.9s	Model setting	○
P0.11	Deceleration time	The time required for VFD to decelerate from maximum output frequency to 0.	0.4~4.0KW 7.5S 5.5~7.5KW 15.0S		
P0.12	Running direction setting	0: forward	0~2	0	○
		1: reverse			
		2: Reverserotationisprohibited			
P0.13	V/F curve setting	0: linearcurve	0~2	0	X
		1: squarecurve			
		2: Multi-pointVFcurve			
P0.14	Torque increase	The manual torque increase is set as a percentage relative to the rated voltage of the motor. If F0.14=0.0, it is vector control.	0.0~30.0%	Model setting	○
P0.15	Torque increase cut-off frequency	This setting is the cut-off frequency point for manual torque increase	0.0~50.0Hz	15.0Hz	X
P0.16	Carrier frequency setting	For situations where silent operation is required, the carrier frequency can be appropriately increased, but increasing the carrier frequency will increase the heat generation of the VFD.	2.0~16.0KHz 0.4~3.0KW 4.0KHz 4.0~7.5KW 3.0KHz	Model setting	X

P0 group - Basic operating parameters

Function code	Name	Content	Range	Factory setting	Change
P0.17	V/F frequency value F1		0.1~frequency value V2	12.5Hz	X
P0.18	V/F voltage value V1		0.0~voltage value F2	25.0%	X
P0.19	V/F frequency value F2		frequency value F1~frequency value F3	25.0Hz	X
P0.20	V/F voltage value V2		voltage value V1~voltage value V3	50.0%	X
P0.21	V/F frequency value F3		frequency value V2~Rated frequency of motor[P4.03]	37.5Hz	X
P0.22	V/F voltage value V3		voltage value V2~100.0%Rated voltage of motor[P4.00]	75.0%	X
P0.23	User password	Setting any non-zero number requires waiting for 3 minutes or power failure to take effect.	0~9999	0	X
P0.24	Frequency display resolution selection	0: 0.1Hz, 1: 1Hz. Pay attention to setting this parameter, and be sure to check the relevant parameters such as maximum output frequency (P0.04), frequency upper limit (P0.05), and motor rated frequency (P4.03).	0~1	0	○
P0.25	Motor control mode	0: V/F control 1: Advanced V/F control 2: Simple Vector Control 3: Advanced Vector Control 4: Torque control	0~4	0	X

P1 group - Auxiliary operating parameters

Function code	Name	Content	Range	Factory setting	Change
P1.00	Starting method	LED ones place: Startup method 0: Start from startup frequency. 1: First apply DC braking and then start from the starting frequency	0000~0011	00	X
		LED ten: Power outage or abnormal restart method 0: invalid 1: Start from startup frequency			
		LED hundreds place: retain			
		LED thousandth place: retain			
P1.01	Startup frequency		0.0~50.0Hz	37.5Hz	O
P1.02	Start DC braking voltage		0.0~50.0% x Rated voltage of motor	0.0%	O
P1.03	Start DC braking time		0.0~30.0s	0.0s	O
P1.04	Stop method	0: decelerate to a stop 1: free stop	0~1	0	X
P1.05	Stop DC braking start frequency		0~Upper limit frequency	0	O
P1.06	Shutdown DC braking voltage		0.0~50.0% x Rated voltage of motor	0.0%	O
P1.07	DC braking time at stop		0.0~30.0s	0.0s	X
P1.08	DC brake waiting time		0.00~99.99s	0.00s	X
P1.09	Positive rotation jog frequency setting	Set the frequency of forward and reverse rotation at a fixed point	0.0~50.0Hz	10.0Hz	O
P1.10	Reverse jog frequency setting				
P1.11	Jogging acceleration time	Set jog acceleration and deceleration time	0.1~999.9S 0.4~4.0KW 10.0S	Model setting	O
P1.12	Jogging deceleration time		5.5~7.5KW 15.0S		

P0 group - basic operating parameters

Function code	Name	Content	Range	Factory setting	Change
P1.13	Jump frequency	By setting the jumping frequency and range, VFD can avoid the mechanical resonance point of the load	0.0~upper frequency limit	0.0Hz	○
P1.14	Jump range		0.0~10.0Hz	0.0Hz	○
P1.15	Frequency combination given method	0: Potentiometer+Digital Frequency 1	0~9	0	X
		1: Potentiometer+Digital Frequency 2			
		2: Potentiometer+AVI			
		3: Digital Frequency 1+AVI			
		4: Digital Frequency 2+AVI			
		5: Digital Frequency 1+multi-speed			
		6: Digital Frequency 2+multi-speed			
		7: Potentiometer+multi-stage speed			
		8: AVI+PLC (in-phase superposition)			
		9: ACI+PLC (in-phase superposition)			
P1.16	Programmable operation control (simple PLC operation)	LED ones place: PLC enabled control 0: invalid, 1: effective	0000~1221	0000	X
		LED ten: Operation mode selection 0: single cycle 1: continuous cycle 2: Maintain the final value after a single cycle			
		LED hundreds: Starting method 0: Restart from the first paragraph onwards 1: Starting from the stage of shutdown (malfunction) moment 2: Start from the stage frequency at the moment of shutdown (malfunction)			
		LED thousandth place: Power down storage selection 0: do not store, 1: store			
P1.17	Multi speed frequency 1	Set segment speed 1 frequency	-Upper limit frequency~Upper limit frequency	5.0Hz	○
P1.18	Multi speed frequency 2	Set segment speed 2 frequency	-Upper limit frequency~Upper limit frequency	10.0Hz	○

P1 group - Auxiliary operating parameters

Function code	Name	Content	Range	Factory setting	Change
P1.19	Multi speed frequency 3	Set segment speed 3 frequency	-Upper limit frequency~Upper limit frequency	15.0Hz	○
P1.20	Multi speed frequency 4	Set segment speed 4 frequency	-Upper limit frequency~Upper limit frequency	20.0Hz	
P1.21	Multi speed frequency 5	Set segment speed 5 frequency	-Upper limit frequency~Upper limit frequency	25.0Hz	
P1.22	Multi speed frequency 6	Set segment speed 6 frequency	-Upper limit frequency~Upper limit frequency	37.5Hz	
P1.23	Multi speed frequency 7	Set segment speed 7 frequency	-Upper limit frequency~Upper limit frequency	50.0Hz	
P1.24	Phase 1 running time	Set segment speed 1 running time (unit selected by [P1.35], default is seconds)	0.0~999.9s	10.0s	○
P1.25	Phase 2 running time	Set segment speed 2 running time (unit selected by [P1.35], default is seconds)	0.0~999.9s	10.0s	○
P1.26	Phase 3 running time	Set segment speed 3 running time (unit selected by [P1.35], default is seconds)	0.0~999.9s	10.0s	○
P1.27	Phase 4 running time	Set segment speed 4 running time (unit selected by [P1.35], default is seconds)	0.0~999.9s	10.0s	○
P1.28	Phase 5 running time	Set segment speed 5 running time (unit selected by [P1.35], default is seconds)	0.0~999.9s	10.0s	○
P1.29	Phase 6 running time	Set segment speed 6 running time (unit selected by [P1.35], default is seconds)	0.0~999.9s	10.0s	○
P1.30	Phase 7 running time	Set segment speed 7 running time (unit selected by [P1.35], default is seconds)	0.0~999.9s	10.0s	○

P0 group - basic operating parameters

Function code	Name	Content	Range	Factory setting	Change
P1.31	Stage acceleration and deceleration time selection 1	LED ones place: Stage 1 Acceleration and Deceleration Time 0-1	0000~1111	0000	○
		LED ten place: Stage 2 Acceleration and Deceleration Time 0-1			
		LED hundreds place: Stage 3 Acceleration and Deceleration Time 0-1			
		LED thousandth place: Stage 4 Acceleration and Deceleration Time 0-1			
P1.32	Stage acceleration and deceleration time selection 2	LED ones place: Stage 5 Acceleration and Deceleration Time 0-1	000~111	000	○
		LED ten place: Stage 6 Acceleration and Deceleration Time 0-1			
		LED hundreds place: Stage 7 Acceleration and Deceleration Time 0-1			
		LED thousandth place: retain			
P1.33	Acceleration time 2	Set acceleration and deceleration time 2	0.0~999.9S 0.4~4.0KW 10.0S 5.5~7.5KW 15.0S	10.0S	○
P1.34	Deceleration time 2				
P1.35	Time unit selection	LED ones place: Process PID time unit	000~211	000	X
		LED ten: Simple PLC time unit			
		LED hundreds: Conventional acceleration and deceleration time unit			
		LED thousandth: retain 0: Unit is 1second 1: Unit is 1minute 2: Unit is 0.1second			

P2 group - Analog and digital input/output parameters					
Function code	Name	Content	Range	Factory setting	Change
P2.00	AVI input lower limit voltage	Set AVI upper and lower limit voltage	0.00~[P2.01]	0.00V	○
P2.01	AVI input upper limit voltage		[P2.01]~10.00v	10.00V	○
P2.02	AVI lower limit corresponding setting	Set the corresponding upper and lower limits of AVI, which corresponds to the percentage of the upper frequency limit [P0.05]	-100.0%~100.0 %	0.0%	○
P2.03	AVI upper limit corresponding setting			100.0%	○
P2.04	ACI input lower limit voltage	Set ACI upper and lower limits of current	0.00~[P2.05]	0.00mA	○
P2.05	ACI input upper limit voltage		[P2.04]~20.00mA	20.00mA	○
P2.06	ACI lower limit corresponding setting	Set the ACI upper and lower limits corresponding setting which is corresponding to the percentage of the upper limit frequency[P0.05]	-100.0%~100.0%	0.0%	○
P2.07	ACI upper limit corresponding setting			100.0%	○
P2.08	Simulated input signal filtering time constant	This parameter is used for filtering the input signals of AVI and panel potentiometers to eliminate the influence of interference	0.1~5.0s	0.1s	○
P2.09	Simulation input anti shake deviation limit	When the analog input signal frequently fluctuates near the given value, the frequency fluctuation caused by this fluctuation can be suppressed by setting P2.09.	0.00~0.10V	0.00V	○
P2.10	Function selection of AO analog output terminal	0: output frequency	0~5	0	○
		1: output current			
		2: motor speed			
		3: output voltage			
		4: AVI			
		5: retain			
P2.11	AO output lower limit	Set the upper and lower limits of AO output	0.00~10.00V/ 0.00~20.00mA	0.00V	○
P2.12	AO output upper limit			10.00V	○

P2 group - Analog and digital input/output parameters

Function code	Name	Content	Range	Factory setting	Change
P2.13	Input terminal X1 function	0: Control end idle 1: Forward jog control 2: Reverse jog control 3: Forward rotation control (FWD) 4: Reverse control (REV) 5: Three line operation control 6: Free shutdown control	0~60	3	X
P2.14	Input terminal X2 function	7: External shutdown signal input (STOP) 8: External reset signal input (RST) 9: External fault normally open input 10: Frequency increment instruction (UP)	0~60	4	X
P2.15	Input terminal X3 function	11: Frequency Decrease Command (DOWN) 13: Multi speed selection S1 14: Multi speed selection S2 15: Multi speed selection S3 16: Force the command channel to be a terminal	0~60	0	X
P2.16	Input terminal X4 function	17: Run command channel to force communication 18: Shutdown DC brake command 19: Switch frequency to AVI 20: Switch frequency to digital frequency 1	0~60	0	X
P2.17	Input terminal X5 function	21: Frequency switch to digital frequency 2 22: Reserved 23: Counter reset signal 24: Counter trigger signal 25: Timer reset signal 26: Timer trigger signal	-0~60	22	X
P2.18	FWD/REV terminal control mode	27: Acceleration and deceleration time selection 28: Pause the swing frequency (stop at the current frequency) 29: Swing frequency reset (return to center frequency) 30: External shutdown/reset signal input (STOP/RST) 31~60: Reserved	0~5	0	X

P2 group - Analog and digital input/output parameters

Function code	Name	Content	Range	Factory setting	Change
P2.19	Terminal function testing selection during power on	The terminal operation command is invalid when powered on The terminal operation command is valid when powered on	0~1	0	X
P2.20	R output setting	0: Idle 1: VFD ready for operation 2: VFD is running 3: VFD is running at zero speed 4: External malfunction shutdown 5: VFD malfunction 6: Frequency/Speed Arrival Signal (FAR) 7: Frequency/speed level detection signal (FDT)	0~14	5	O
P2.21	Y1 output setting	8: Output frequency reaches the upper limit 9: The output frequency has reached the lower limit 10: VFD overload warning 11: Timer overflow signal 12: Counter detection signal 13: Counter reset signal 14: Auxiliary motor	0~14	0	◆
P2.22	R closure delay	Delay from relay R state change to output change	0.0~255.0s	0.0s	X
P2.23	R disconnection delay				
P2.24	Frequency reaches FAR detection amplitude	The output frequency is within the positive and negative detection width of the set frequency, and the terminal outputs a valid signal (low level)	0.0Hz~15.0Hz	5.0Hz	O
P2.25	FDT level setting value		0.0Hz~upper frequency limit	10.0Hz	O
P2.26	FDT lag value		0.0~30.0Hz	1.0Hz	O

P2 group - Analog and digital input/output parameters

Function code	Name	Content	Range	Factory setting	Change
P2.27	UP/DOWN terminal modification rate	This function code is used to set the frequency change rate of the UP/DOWN terminal when setting the frequency, that is, the magnitude of the frequency change when the UP/DOWN terminal is short circuited to the COM terminal for one second.	0.1Hz~99.9Hz/s	1.0Hz/1s	○
P2.28	Setting of input terminal pulse triggering mode (M1~M5)	0: indicates the level triggering method 1: Indicate the pulse triggering method	0~1FH	0	○
P2.29	Effective logic settings for input terminals (M1~M5)	0: indicates positive logic, that is, Mi terminal is connected to the common terminal effectively, and disconnected is invalid 1: Expressing anti logic, that is, the connection between Mi terminal and common terminal is invalid, while disconnection is valid	0~1FH	0	○
P2.30	X1 filtering coefficient	Used to set the sensitivity of input terminals. If the digital input terminal is susceptible to interference and causes misoperation, this parameter can be increased to resist The interference capability is enhanced, but setting it too high will result in a decrease in the sensitivity of the input terminal. 1: Represents a 2MS scanning time unit.	0~9999	5	○
P2.31	X2 filtering coefficient		0~9999	5	○
P2.32	X3 filtering coefficient		0~9999	5	○
P2.33	X4 filtering coefficient		0~9999	5	○
P2.34	X5 filtering coefficient		0~9999	5	○

P3 group - PID parameters

Function code	Name	Content	Range	Factory setting	Change
P3.00	PID function setting	LED ones place: PID regulation characteristics 0: Invalid 1: Positive effect: When the feedback signal is greater than the given amount of PID, the VFD output frequency is required to decrease (i.e. reduce the feedback signal).	0000~2122	1010	X
		LED ten place: PID input channel for given quantity 0: Keyboard potentiometer, PID given quantity is given by the potentiometer on the operation panel. 1: Number given, PID given quantity is given by number and set by function code P3.01. 2: Pressure given (MPa, Kg) Set the pressure to P3.01 and P3.18.			
		LED Hundred place: PID Feedback Input Channel 0: AVI 1: ACI			
		LED Thousand place: PID Sleep Selection 0: Invalid 1: Normal hibernation. This method requires setting specific parameters such as P3.10~P3.13. 2: The parameter settings for disturbance sleep and sleep mode selection are the same when 0 is selected. If the PID feedback value is within the range of P3.14 setting, disturbance sleep will be entered after maintaining the sleep delay time. When the feedback value is less than the wake-up threshold (PID polarity is positive), immediately wake up.			

P3 group - PID parameters

Function code	Name	Content	Range	Factory setting	Change
P3.01	Set quantitative numbers	Set the given amount of PID control using the operation keyboard. This function is only effective when the PID given channel is selected with a given number (P3.00 ten digit is 1 or 2). If the decimal place of P3.00 is 2, it is used as a pressure setting, and this parameter is consistent with the unit of P3.18.	0.0~100.0%	0.0%	○
P3.02	Feedback channel gain	When the feedback channel is not at the same level as the set channel, this function can be used to adjust the gain of the feedback channel signal.	0.01~10.00	1.00	○
P3.03	Proportional gain P	The speed of PID regulation is set through two parameters: proportional gain and integration time. If the regulation speed is fast, the proportional gain needs to be increased and the integration time needs to be reduced. If the regulation speed is slow, the proportional gain needs to be reduced and the integration time needs to be increased. In general, the differential time is not set.	0.01~5.00	2.00	○
P3.04	Integral time Ti		0.1~50.0s	1.0s	○
P3.05	Differential time Td		0.1~10.0s	0.0s	○
P3.06	Sampling period T	The larger the sampling period, the slower the response, but the better the suppression effect on interference signals, so it is generally not necessary to set it.	0.1~10.0s	0.0s	○
P3.07	Deviation limit	The deviation limit is the ratio of the absolute value of the deviation between the system feedback quantity and the given quantity to the given quantity. When the feedback quantity is within the deviation limit range, the PID regulation does not act.	0.0~20.0%	0.0%	○
P3.08	Closed loop preset frequency	The frequency and running time of the frequency converter before the PID is put into operation	0.0~upper frequency limit	0.0Hz	○
P3.09	Preset frequency holding time		0.0~999.9s	0.0s	X

P0 group - basic operating parameters

Function code	Name	Content	Range	Factory setting	Change
P3.10	Awakening threshold coefficient	If the actual feedback value is greater than the set value and the frequency output by the VFD reaches the lower limit frequency, the VFD enters sleep mode after a delay waiting time defined in P3.12; This value is a percentage of the PID set value.	0.0~150.0%	100.0%	○
P3.11	Awakening threshold coefficient	If the actual feedback value is less than the set value, the VFD will leave sleep mode and start working after the delay waiting time defined in P3.13; This value is a percentage of the PID set value.	0.0~150.0%	90.0%	○
P3.12	Sleep delay time	Set sleep delay time	0.0~999.9s	100.0s	○
P3.13	Wake up delay time	Set wake-up delay time	0.0~999.9s	1.0s	○
P3.14	Deviation between feedback during sleep and set pressure	This function parameter is only valid for disturbance sleep mode	0.0~10.0%	0.5%	○
P3.15	Delay time for tube burst detection	Set the delay time for tube burst detection	0.0~130.0s	0.0s	○
P3.16	High pressure detection threshold	When the feedback pressure is greater than or equal to this set value, a burst fault "EPA0" will be reported after a P3.15 burst delay. When the feedback pressure is less than this set value, the burst fault "EPA0" will automatically reset; This threshold is a percentage of the given pressure.	0.0~200.0%	150.0%	○
P3.17	Low pressure detection threshold	When the feedback pressure is less than this set value, a burst fault "EPA0" will be reported after a P3.15 burst delay. When the feedback pressure is greater than or equal to this set value, the burst fault "EPA0" will automatically reset; This threshold is a percentage of the given pressure.	0.0~200.0%	50.0%	○
P3.18	Sensor range	Set the maximum range of the sensor	0.00~99.99 (Mpa,Kg)	10.00Mpa	○

P4 group - Advanced functional parameters

Function code	Name	Content	Range	Factory setting	Change
P4.00	Rated voltage of motor	Motor parameter settings	0~500V:380V 0~250V:220V	Model setting	X
P4.01	Rated current of motor		0.1~999.9A	Model setting	X
P4.02	Rated speed of motor		0~60000Krpm	Model setting	X
P4.03	Rated frequency of motor		1.0~999.9Hz	50.0Hz	X
P4.04	Motor stator resistance	Set the stator resistance of the motor	0.001~20.000Ω	Model setting	O
P4.05	Motor no-load current	Set the no-load current of the motor	0.1~[P4.01]	Model setting	X
P4.06	AVR function	0: Invalid	0~2	0	X
		1: Effective throughout the entire process			
		2: Invalid only when decelerating			
P4.07	Cooling fan control	0: Automatic control mode	0~1	0	O
		1: The power on process runs continuously			
P4.08	Number of automatic fault resets	When the number of obstacle resets is set to 0, there is no automatic reset function and only manual reset can be performed. 10 means there is no limit to the number of times, that is, countless times.	0~10	0	X
P4.09	Automatic fault reset interval time	Set the interval time for automatic fault reset	0.5~25.0s	3.0s	X
P4.10	Starting voltage of energy consumption braking	If the DC side voltage inside the VFD is higher than the starting voltage for energy consumption braking, the built-in braking unit will activate. If a braking resistor is connected at this time, the increased voltage energy inside the VFD will be released through the braking resistor, causing the DC voltage to fall back.	330~380V/ 660V~800V	385/720	O
P4.11	Energy consumption braking action ratio		10~100%	100%	O
P4.12	Overmodulation function selection	0: Invalid, 1: Valid	0~1	0	X
P4.13	PWM mode	0: Full frequency seven band	0~3	0	X
		1: Full frequency five band			
		2: Seven sections to five sections			
		3: Unidirectional driving mode			

P4 group - Advanced functional parameters

Function code	Name	Content	Range	Factory setting	Change
P4.14	Slip compensation coefficient	When an asynchronous motor is loaded, it will cause a decrease in speed. Using slip compensation can make the motor speed approach its synchronous speed, thereby increasing the accuracy of motor speed control. This coefficient is invalid for ordinary V/F and simple vectors.	0~200%	100%	X
P4.15	Slip compensation mode	0: Invalid	0~1	0	X
		1: Low frequency compensation, please note that this parameter is only valid for advanced V/F			
P4.16	Self learning of motor parameters	0: Invalid	0~1	0	X
		1: Static self-learning (STAR displayed immediately upon startup, END displayed upon termination, and extinguished after 1 second)			
P4.17	Motor rated power	After changing the rated power of the motor P4.17, P4.01, P4.02, P4.04, P4.05, P4.18~P4.20 will automatically update to the default parameters of the corresponding motor power.	0.0~2000.0KW	Model setting	○
P4.18	Motor rotor resistance		0.00~200.00Ω	Model setting	○
P4.19	Motor stator and rotor inductance		0.00~200.00mH	Model setting	○
P4.20	Motor stator and rotor mutual inductance		0.00~200.00mH	Model setting	○
P4.21	Speed loop (ASR1) proportional gain	Function codes P4.21~P4.26 are valid in vector control mode. By setting the proportional gain P and integration time I, the speed response characteristics of vector control can be changed.	1~100	30	X
P4.22	Speed Loop (ASR1) integration time		0.01~10.00s	0.50	○
P4.23	Switching low point frequency		0.0~10.0Hz	5.0	X
P4.24	Speed loop (ASR2) proportional gain		1~100	20	○
P4.25	Speed Loop (ASR2) integration time		0.01~10.00s	1.00	○
P4.26	Switch high frequency		[P4.23]~320.0Hz	10.0	X

P4 group - Advanced functional parameters

Function code	Name	Content	Range	Factory setting	Change
P4.27	Vector slip compensation	In vector control mode, this parameter is used to adjust the accuracy of the motor's stable speed. When the motor is overloaded and the speed is low, this parameter is increased; otherwise, it is decreased.	50%~200%	100	○
P4.28	Speed loop filtering time constant	Set the filtering time constant for the speed loop	0.000~1.000s	0.010	○
P4.29	Retain	-	-	0	◆
P4.30	Speed loop torque limit	The percentage of the rated current of the motor at this set value	0.0%~200.0%	150.0	○
P4.31	Torque command selection	0: keyboard number given	0~2	0	X
		1: AVI(0~10V/0-20mA)			
		2: Reserved			
P4.32	Torque digital given	The percentage of the rated current of the motor at this set value	0.0%~200.0%* Rated current of motor	150.0	○
P4.33	Maximum frequency of forward torque control	Used to set the maximum forward/reverse operating frequency of VFD under torque control mode	0.0~3200.0Hz	50.0	○
P4.34	Maximum frequency of reverse torque control		0.0~3200.0Hz	50.0	○
P4.35	Torque rise time	The torque rise/fall time defines the time it takes for the torque to rise from 0 to its maximum value or to fall from its maximum value to 0.	0.00~1.00s	0.00	○
P4.36	Torque reduction time		0.00~1.00s	0.00	○

P5 group - Protection function parameters

Function code	Name	Content	Range	Factory setting	Change
P5.00	Protection settings	LED one: motor overload protection selection 0: Invalid 1: Effective	0000~1211	0001	X
		LED ten: PID feedback disconnection protection 0: Invalid 1: Protective action and free shutdown			
		LED Hundred Position: Reserved			
		LED Thousand Position: Vibration Suppression Selection 0: Invalid 1: Effective			
P5.01	Overload protection coefficient of motor	The motor overload protection factor is the percentage of the rated current value of the motor to the rated output current value of the frequency converter.	30%~110%	100%	X
P5.02	Undervoltage protection level	This function code specifies the lower limit voltage allowed for the DC bus when the VFD is working normally.	50~280/ 50~480V	180/360V	X
P5.03	Deceleration voltage limit coefficient	This parameter is used to adjust the ability of VFD to suppress overvoltage during deceleration.	0: Close 1~255	1	X
P5.04	Overvoltage limit level	The overvoltage limit level defines the operating voltage during overvoltage stall protection.	350~400/ 660~850V	375/790V	X
P5.06	Constant speed current limiting coefficient	This parameter is used to adjust the ability of VFD to suppress overcurrent during constant speed processes.	0: Close 1~10	0	X
P5.07	Current limiting level	The current limiting level defines the current threshold for automatic current limiting action, which is set as a percentage relative to the rated current of the frequency converter.	50%~250%	160%	X

P5 group - Protection function parameters

Function code	Name	Content	Range	Factory setting	Change
P5.08	Feedback Disconnection Detection Value	The percentage given by PID, when the feedback value of PID is continuously less than the feedback disconnection detection value, VFD will take corresponding protective actions according to the setting of P5.00, which is invalid when P5.08=0.0%.	0.0~100.0%	0.0%	X
P5.09	Feedback on wire breakage detection time	Delay time before protection action after feedback disconnection occurs.	0.1~999.9s	10.0s	X
P5.10	Overload warning level of frequency converter	The current threshold for VFD overload warning action is set as a percentage relative to the rated current of VFD.	0~150%	120%	O
P5.11	VFD overload warning delay	The delay time between the VFD output current continuously exceeding the overload pre alarm level amplitude (P5.10) and the output overload pre alarm signal.	0.0~9999.0s	5.0s	X
P5.12	Activate jog priority	0: Invalid	0~1	0	X
		1: When the frequency converter is running, jog has the highest priority			
P5.13	Oscillation suppression coefficient	When there is motor oscillation, it is necessary to set P5.00 thousand effective, turn on the oscillation suppression function, and then adjust it by setting the oscillation suppression coefficient. Generally, if the oscillation amplitude is large, increase the oscillation suppression coefficient P5.13, P5.14-P5.16 does not need to be set; If encountering special occasions, P5.13 to P5.16 should be used together.	0~200	30	O
P5.14	Amplitude suppression coefficient		0~12	5	O
P5.15	Oscillation suppression lower limit frequency		0.0~[P5.16]	5.0Hz	O
P5.16	Oscillation suppression upper limit frequency		[P5.15]~[P0.05]	45.0Hz	O
P5.17	Wave by wave current limiting selection	LED one: selected for acceleration 0: Invalid, 1: Effective	000~111	011	X
		LED ten: Slow down Selection 0: Invalid, 1: Effective			
		LED hundred: Constant Speed Selection 0: Invalid, 1: Effective			
		LED thousand: Reserved			

P6 group - Communication parameters

Function code	Name	Content	Range	Factory setting	Change
P6.00	Local address	Set local address, 0 is the broadcast address	0~247	1	X
P6.01	MODBUS communication configuration	LED one: Baud rate selection 0: 9600BPS, 1: retain, 2: retain	0000~0322	0001	X
		LED ten: data format 0: No, 1: even parity, 2: odd parity			
		LED hundred: Communication response method 0: Normal, 1: Only respond to slave addresses, 2: No response 3: The slave does not respond to the free stop command of the host in broadcast mode			
		LED thousand: Reserved			
P6.02	Communication timeout detection time	If the local machine does not receive the correct data signal within the time interval defined by this function code, it will consider that there is a communication failure, and the VFD will decide whether to protect or maintain the current operation according to the setting of the communication failure action mode. At 0.00, do not perform RS485 communication timeout detection	0.1~100.0s	0.0s	X
P6.03	Local response delay	The intermediate time interval between receiving VFD data frames and sending response data frames to the upper computer. If the response time is less than the system processing time, the system processing time shall prevail.	0~200ms	5ms	X
P6.04	Proportional linkage coefficient	Set VFD as the weight coefficient for the frequency command received by the slave through the RS485 interface, and the actual operating frequency is equal to the product of this function code value and the frequency setting command value received through the RS485 interface. In linkage control, this function code can set the ratio of operating frequencies for multiple VFDs.	0.01~10.00	1.00	O

P7 group - Supplementary functional parameters

Function code	Name	Content	Range	Factory setting	Change
P7.00	Counting and timing mode	LED one: counting arrival processing 0: Weekly count, stop output 1: Single week count, continue to output 2: Cycle count, stop output 3: Cycle counting, continue outputting	000~303	103	X
		LED ten: reserved			
		LED hundred: timed arrival processing 0: Weekly timed, stop output 1: Weekly timer, continue to output 2: Cycle timing, stop output 3: Cycle timing, continue outputting			
		LED thousand: Reserved			
P7.01	Counter reset value setting	Set the counter reset value	[P7.02]~9999	1	O
P7.02	Counter detection value setting	Set counter detection value	0~[P7.01]	1	O
P7.03	Timed time setting	Set timer time	0~9999s	0	O
P7.04~P7.07	Retain	-	-	0	O
P7.08	Swing frequency control	0: Prohibited	0~1	0	X
		1: Effective			
P7.09	Swing control	0: Fixed amplitude, with the amplitude reference value being the maximum output frequency (P0.04)	0~1	0	X
		1: Variable amplitude, with the amplitude reference value being the given communication frequency.			
P7.10	Selection of start-up mode for swing frequency shutdown	0: Start according to the memory state before shutdown	0~1	0	X
		1: Restart			

P7 group - Supplementary functional parameters

Function code	Name	Content	Range	Factory setting	Change
P7.11	Swing control	The swing amplitude is a percentage relative to the maximum output frequency (P0.04)	0.0~100.0%	0.0%	○
P7.12	Jumping frequency	This function code refers to the amplitude of rapid decrease when the frequency reaches the upper limit of the swing frequency during the swing frequency process, and of course, it also refers to the amplitude of rapid increase when the frequency reaches the lower limit of the swing frequency. This refers to the percentage relative to the amplitude of the swing frequency (P7.11), set to 0.0%, which means there is no sudden jump frequency.	0.0~50.0%	0.0%	○
P7.13	Swing frequency rise time	The running time from the lower limit frequency to the upper limit frequency of the swing frequency	0.1~3600.0s	5.0	○
P7.14	Swing frequency descent time	The running time from the upper limit frequency to the lower limit frequency of the swing frequency	0.1~3600.0s	5.0	○
P7.15	Upper limit frequency delay of swing frequency	Set the upper and lower limit frequency delay for swing frequency	0.1~3600.0s	5.0	○
P7.16	Lower limit frequency delay of swing frequency		0.1~3600.0s	5.0	○

P8 group - Manage and display parameters

Function code	Name	Content	Range	Factory setting	Change
P8.00	Selection of Monitoring Parameters for Operation	For example, P8.00=2, which means selecting the output voltage (d-02), the default display item on the main monitoring interface is the current output voltage value.	0~31	0	○
P8.01	Selection of shutdown monitoring parameter items	For example, P8.01=3, which means selecting the bus voltage (d-03), the default display item on the main monitoring interface is the current bus voltage value.	0~31	1	○
P8.02	Motor speed display coefficient	Used to correct the display error of the speed scale, without affecting the actual speed.	0.01~99.99	1.00	○
P8.03	Parameter initialization	0: No operation VFD is in normal parameter read and write state. Function code setting value. Whether it can be changed depends on the setting status of the user password and the current working status of VFD.	0~2	0	X
		1: Restore factory settings All user parameters are reset to factory settings according to the model.			
		2: Clear fault records Perform a reset operation on the contents of fault records (d-19~d-24). After the operation is completed, this Function code automatically clears 0.			
P8.04	MF key setting	0: JOG	0~3	0	X
		1: Forward and reverse switching			
		2: Clear ▲/▼ key frequency setting			
		3: Reverse operation (at this time, the RUN key defaults to forward rotation)			

P8 group - Manage and display parameters

Function code	Name	Content	Range	Factory setting	Change
P8.05	Slip compensation selection	0: Invalid	0~1	0	X
		1: Effective When an asynchronous motor is loaded, it will cause a decrease in speed. Using slip compensation can make the motor speed approach its synchronous speed, thereby increasing the accuracy of motor speed control.			
P8.06	Instantaneous power failure frequency reduction coefficient	Set the instantaneous power down frequency reduction coefficient, the larger the setting value, the faster the frequency reduction.	0: The instant stop function is invalid 1~9999	1	O
P8.07	Instantaneous power failure and frequency reduction point	Set instantaneous power down and frequency reduction point	220V:180~330V 250V 380V:300~550V 450V	Model setting	X
P8.08	MPPT low point operating voltage	If the bus voltage (d-03) is higher than the set value of the MPPT high point operating voltage (P8.09), operate at maximum frequency; If the bus voltage (d-03) is lower than the set value of MPPT high point operating voltage (P8.09), operate at the frequency obtained	0V~[F1.00]	Model setting	O
P8.09	MPPT high point operating voltage	by multiplying (bus voltage/MPPT high point operating voltage) by the maximum frequency. If the bus voltage reaches the MPPT low point operating voltage (P8.08), operate at the lowest operating frequency of the outlet (P8.11).	[F1.00]~1000V	Model setting	O

P8 group - Manage and display parameters

Function code	Name	Content	Range	Factory setting	Change
P8.10	Proportion of no-load current corresponding to water shortage detection current of photovoltaic water pump	If the VFD operates above the minimum outlet frequency and the output current is less than the motor no-load current (P4.05)*The proportion of no-load current corresponding to the water shortage detection current of the photovoltaic water pump (P8.10). After the water shortage detection time of the photovoltaic water pump (P8.10), the VFD reports a water shortage fault E-24	0.0~300.0%	0.0	○
P8.11	Minimum operating frequency of photovoltaic water pump effluent		0.0~999.9Hz	0.0	○
P8.12	Photovoltaic water pump water shortage detection time		0~250s	10	○
P8.13	Running auxiliary display monitoring parameter project	For example, P8.00=2, which means selecting the output voltage (d-02). So the default display item on the main monitoring interface is the current output voltage value.	0~31	4	○
P8.14	Selection of Monitoring Parameters for Shutdown Auxiliary Display	For example, P8.01=3, which means selecting the bus voltage (d-03). So the default display item on the main monitoring interface is the current bus voltage value.	0~31	3	○

P9 group - Factory parameters

Function code	Name	Content	Range	Factory setting	Change
P9.00	Factory password	1~9999	1	****	◇

d group - Monitoring parameter group

Function code	Name	Content	Range	Factory setting	Change
d-00	Output frequency (Hz)	0.0~999.9Hz	0.1Hz	0.0Hz	◆
d-01	Set frequency (Hz)	0.0~999.9Hz	0.1Hz	0.0Hz	◆
d-02	Output voltage (V)	0~999V	1V	0V	◆
d-03	Bus voltage (V)	0~999V	1V	0V	◆
d-04	Output current (A)	0.0~999.9A	0.1A	0.0A	◆
d-05	Motor speed (Krpm)	0~60000Krpm	1Krpm	Model setting	◆
d-06	Analog input AVI(V/mA)	0.00~10.00V/0.00~20.00mA	0.01V	0.00V/mA	◆
d-07	Analog input ACI (mA)	0.00~20.00mA	0.01mA	0.00mA	◆
d-08	Analog output AO (V/mA)	0.00~10.00V/0.00~20.00mA	0.01V/0.01mA	0.00V/mA	◆
d-09	Reserved	-	-	0	◆
d-10	Reserved	-	-	0	◆
d-11	PID pressure feedback value	0.00~10.00V/0.00~99.99(Mpa.Kg)	0.01V/(Mpa.Kg)	0.00V/(Mpa.Kg)	◆
d-12	Current count value	0~9999s	1s	0s	◆
d-13	Current timer value (s)	0~9999s	1s	0s	◆
d-14	Input terminal status (X1-X5)	0~1FH	1H	0H	◆
d-15	Output relay status (R)	0~1H	1H	0H	◆
d-16	Module temperature (°C)	0.0~132.3°C	0.1°C	0.0	◆
d-17	Software upgrade date (year)	From 2010 to present	1	2017	◆
d-18	Software upgrade date (month, day)	0~1231	1	0914	◆
d-19	Second fault code	0~19	1	0	◆
d-20	Last fault code	0~19	1	0	◆

P8 group - Manage and display parameters

Function code	Name	Content	Range	Factory setting	Change
d-21	Output frequency (Hz) at the latest fault	0.0~999.9Hz	0.1Hz	0.0Hz	◆
d-22	Output current at the most recent fault (A)	0.0~999.9A	0.1A	0.0V	◆
d-23	Bus voltage (V) at the most recent fault	0~999V	1V	0V	◆
d-24	Module temperature at the most recent fault (°C)	0.0~132.3°C	0.1°C	0.0°C	◆
d-25	VFD cumulative running time (h)	0~9999h	1h	0h	◆
d-26	VFD status	0 to FFFFH	1H	0H	◆
		BIT0: Run/Stop			
		BIT1: Reverse / Forward			
		BIT2: Jog			
		BIT3: DC braking			
		BIT4: Reserved			
		BIT5: Overvoltage limit			
		BIT6: Constant speed down frequency			
		BIT7: Overcurrent limit			
		BIT8~9:00-zero speed/01-acceleration/10-deceleration/11-uniform speed			
		BIT10: Overload pre-alarm			
		BIT11: Reserved			
		BIT12~13 running command channel: 00-panel /01-terminal/10-reserved			
		BIT14~15 bus voltage status: 00-normal/01-low voltage protection/10-overpressure protection			
d-27~29	Retain	-			◆
d-30	Output torque	-200~+200%	1%	0%	◆
d-31	Input voltage	0~999V	1V	Model setting	◆

E group - Fault code				
Fault code	Name	Cause of failure	Fault countermeasure	Code name
E0C1	Accelerate overcurrent during operation	Acceleration time is too short	Increase acceleration time	1
		The VFD power is too small	Use a vfd with a large power rating	
		Improper setting of V/F curve or torque boost	Adjust the V/F curve or torque boost	
E0C2	Over-current during deceleration	Deceleration time is too short	Increase deceleration time	2
		The VFD power is too small	Use a frequency converter with a large power rating	
E0C3	Overcurrent in constant speed operation	Low voltage of the power grid	Check the input power supply	3
		Sudden or abnormal load changes	Check the load or reduce the load mutation	
		The frequency inverter power is too small	Use a VFD with a large power rating	
EHU1	Overvoltage during acceleration operation	Abnormal input voltage	Check the input power supply	4
		Restart the rotating motor	Set to start after DC braking	
EHU2	Overpressure during deceleration operation	Deceleration time is too short	Increase deceleration time	5
		Abnormal input voltage	Check input power	
EHU3	Overvoltage during constant speed operation	Abnormal input voltage	Check input power	6
EHU4	Overvoltage during shutdown	Abnormal input voltage	Check the power supply voltage	7
ELU0	Undervoltage during operation	Abnormal input voltage or failure of the starter to engage	Check the power supply voltage	8
ESC1	Functional module malfunction	VFD output short circuit or grounding	Check the motor wiring	9
		VFD instantaneous overcurrent	Reference overcurrent countermeasures	
		Abnormal or severe interference in the control board	Seek help from the manufacturer	
		Power device damage	Seek help from the manufacturer	
E-OH	Radiator overheating	The ambient temperature is too high	Reduce the ambient temperature	10
		Fan damaged	Replace the fan	
		Fan blockage	Ventilation duct	

E group - Fault code

Fault code	Name	Cause of failure	Fault countermeasure	Code name
EOL1	VFD overload	V/F curve or torque boost error	Adjust V/F curve and torque	11
		Low voltage of the power grid	Check voltage	
		The acceleration time is too short	Extend acceleration time	
		Excessive motor load	Choose a high-power VFD	
EOL1	Motor overload	V/F curve or torque boost error	Adjust V/F curve and torque	12
		Low voltage of the power grid	Check voltage	
		Motor blockage or sudden overload change	Check the load	
		Motor overload protection factor setting error	Correctly set the motor overload protection factor	
E-EF	External device malfunction	External device malfunction input terminal closed	Disconnect the faulty input terminal of the external device and remove the fault	13
EPOF	Dual CPU communication failure	CPU communication failure	Seek help from the manufacturer	
EPID	PID feedback disconnection	Loose PID feedback line	Check the feedback line	14
		The feedback amount is less than the wire breakage detection value	Adjust the detection input threshold	
E485	Overpressure during deceleration operation	Mismatch with the baud rate of the upper computer	Adjust baud rate	16
		RS485 channel interference	Check if the communication wiring is shielded.	
		Communication timeout	Retry	
ETUN	Motor tuning fault	Motor parameter setting error	Reset motor parameters	17
ECCF	Current detection fault	Motor sampling circuit fault	Seek help from the manufacturer	18
		Auxiliary power failure		
EEEP	EEPROM read/write error	EEPROM malfunction	Seek help from the manufacturer	19
EPLI	Output phase loss protection	Output U, V, W have phase loss	Check the output wiring	20
E-LT	Photovoltaic water pump water shortage fault	Abnormal water pressure/level	Check if the water pressure at the inlet of the water pump is abnormal	21
		No feedback signal	Check the installation and wiring of the sensor	
EPAO	Explosion tube malfunction	Feedback pressure is less than the low pressure detection threshold or greater than or equal to the detection threshold	Check the feedback connection or adjust the high and low voltage threshold for detection	22

RS485 Communication Protocol

(All data below is in hexadecimal)

1. RTU mode and format

When the controller communicates on the Modbus bus in RTU mode, every 8-bit byte in the information is divided into 4-bit hexadecimal characters. The main advantage of this mode is that the density of transmitted characters is higher than that of ASCII mode at the same baud rate, and each message must be transmitted continuously.

(1) Format of each byte in RTU mode

Coding system: 8-bit binary, hexadecimal 0-9, A-F.

Data bits: 1 start bit, 8 data bits (low priority), 1 stop bit, parity bit is optional. (Reference RTU data frame as sequence diagram)

Error checking area: cyclic redundancy check (CRC)

(2) Frame bitmap of RTU data

with parity check

Start	1	2	3	4	5	6	7	8	Par	Stop
-------	---	---	---	---	---	---	---	---	-----	------

Without parity check

Start	1	2	3	4	5	6	7	8	Stop
-------	---	---	---	---	---	---	---	---	------

2. Description of read and write function code

Function code	Function description
03	Read register
06	Write register

3. Parameter address description of communication

Function	Address description	Data statement	R/W
Communication control command	2000H	0001H: Stop	W
		0012H: Forward	
		0013H: Forward jog	
		0022H: Reverse	
		0023H: Reverse jog	
Address of communication frequency setting	2001H	The communication setting frequency range is -10000~10000 Note: The communication setting frequency is a percentage relative to the maximum frequency, ranging from -10.00% to 100.00%.	W
Communication control command	2002H	0001H: External fault input	W
		0002H: Fault reset	

Function	Adress description	Data statement	R/W
Read fault code instructions	2100H	00: normal	R
		01: Module failure	
		02: Over voltage	
		03: Temperature failure	
		04: VFD overload	
		05: Motor overload	
		06: External failure	
		07-09: Reserved	
		10: Overcurrent in Acceleration	
		11: Overcurrent in deceleration	
		12: overcurrent medium constant speed	
		13: retain	
		14: undervoltage	
Read the instructions for running/stopping parameters	2101H	Bit0: Run	R
		Bit1: Stop	
		Bit2: Jog	
		Bit3: Forward	
		Bit4: Reverse	
		Bit5-7: Retain	
		Bit8: Communication given	
		Bit9: Analog signal input	
		Bit10: Communication operation command	
		Bit11: Parameter locking	
		Bit12: Running	
		Bit13: Jog command	
		Bit14-15: Retain	
	2102H	Set frequency (two decimal places)	R
	2103H	Output frequency (two decimal places)	R
	2104H	Output current (to one decimal place)	R
	2105H	Bus voltage (to one decimal place)	R
	2106H	Output voltage (to one decimal place)	R
	2107H	Analog input AVI (two decimal places)	R
	2108H	Analog input ACI (two decimal places)	R
	2109H	Current count value	R
	210AH	Motor speed	R
	210BH	Analog output AO(two decimal places)	R
	210CH	Retain	R
	210DH	VFD temperature(to one decimal place)	R
	210EH	PID feedback value(two decimal places)	R
	210FH	PID setting value(two decimal places)	R
	2110H	Retain	R

Function	Adress description	Data statement	R/W
Read run /stop parameters description	2111H	Pulse input frequency	R
	2112H	Current fault	R
	2113H	Current timer value	R
	2114H	Input terminal status	R
	2115H	Output terminal status	R
	2116H	BIT0: Run/Stop	R
		BIT1: Forward/Reverse	
		BIT2: Jog	
		BIT3: DC braking	
		BIT4: Retain	
		BIT5: Overvoltage limit	
		BIT6: Constant speed frequency reduction	
		BIT7: Overcurrent restriction	
		BIT8-9: 00- Zero speed/01- Acceleration/10- Deceleration/11- Uniform speed	
		BIT10: Overload pre alarm	
		BIT11: Retain	
		BIT12-13: Run command channel: 00 panel/01 terminal/10 communication	
		BIT14-15: Bus voltage status: 00- normal/01- low voltage protection/10- overvoltage protection	

4. 03 Reading Function Mode

Inquiry information frame for mat

Address	01H
Function	03H
Starting data address	21H
	02H
Data(2Byte)	00H
	02H
CRC CHK Low	6FH
CRC CHK High	F7H

This section of data analysis:

01H-----VFD address

03H-----Read function code

2102H-----As the starting address

0002H-----Read the number of addresses and 2102H and 2103H

F76FH-----16 bit CRC code

Response information frame format:

Address	01H
Function	03H
Data Num*2	04H
Data 1 [2 byte]	17H
	70H
Data 2 [2 byte]	00H
	00H
CRC CHK Low	FEH
CRC CHK High	5CH

This section of data analysis:

01H-----VFD address

03H-----Read function code

04H-----Read term times the product of 2

1770H-----Read data at 2102H (set frequency)

0000H-----Read data at 2103H (output frequency) 5CFEH-----16 bit CRC code

5. 06H Write function mode

Inquiry information frame format

Address	01H
Function	06H
Starting data address	20H
	00H
Data [2 byte]	00H
	01H
CRC CHK Low	43H
CRC CHK High	CAH

This section of data analysis:

01H-----VFD address

06H-----Write function code

2000H-----Control command address

0001H-----Stop command

43CAH-----16 bit CRC code

Response information frame format:

Address	01H
Function	06H
Starting data address	20H
	00H
Number of Data(Byte)	00H
	01H
CRC CHK Low	43H
CRC CHK High	CAH

This section of data analysis: returns the same input data if set correctly.